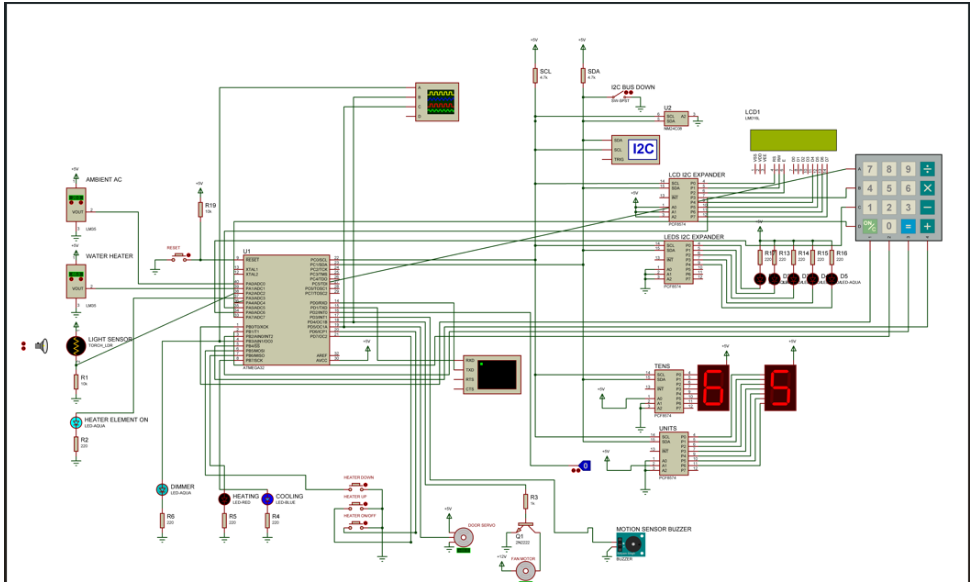
GRADUATION PROJECT

ATmega32 Smart Home + Electric Water Heater

Abdelrahman Sherif Medhat

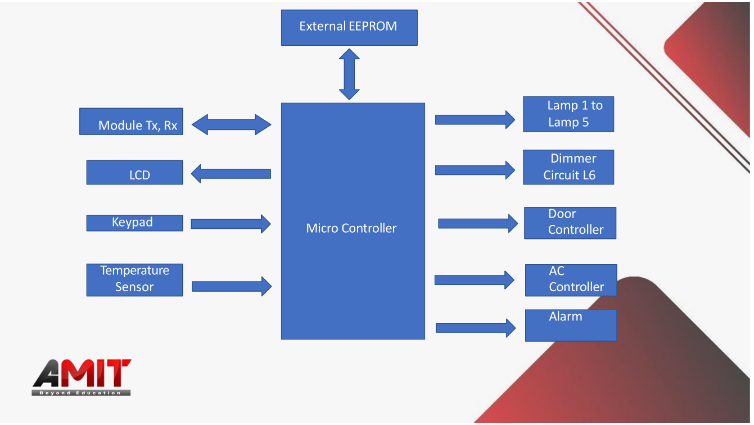
Embedded Systems Diploma · AMIT Learning

Instructor: Eng. Eslam Hefny



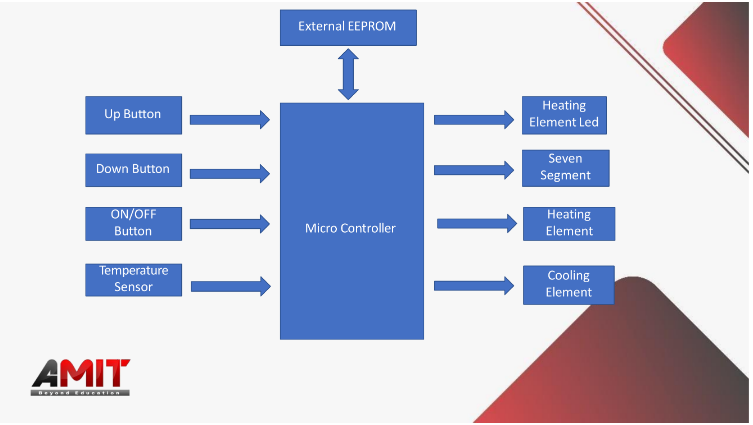
SCOPE

Two course projects, one MCU



Project 1 · Smart Home

+

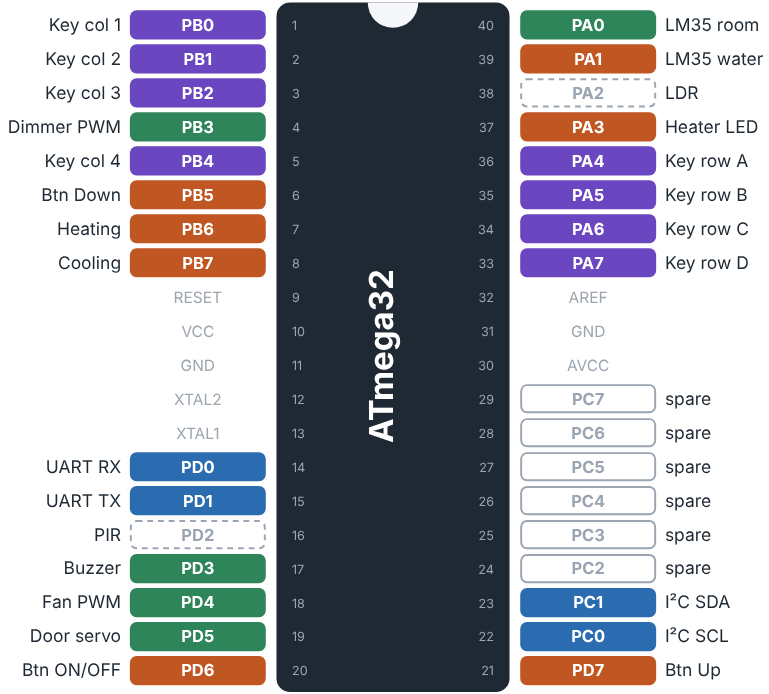


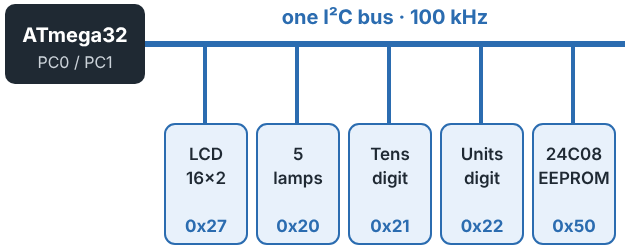
Project 2 · Electric Water Heater

The heater becomes one appliance of the smart home

PIN BUDGET

One MCU, 32 pins



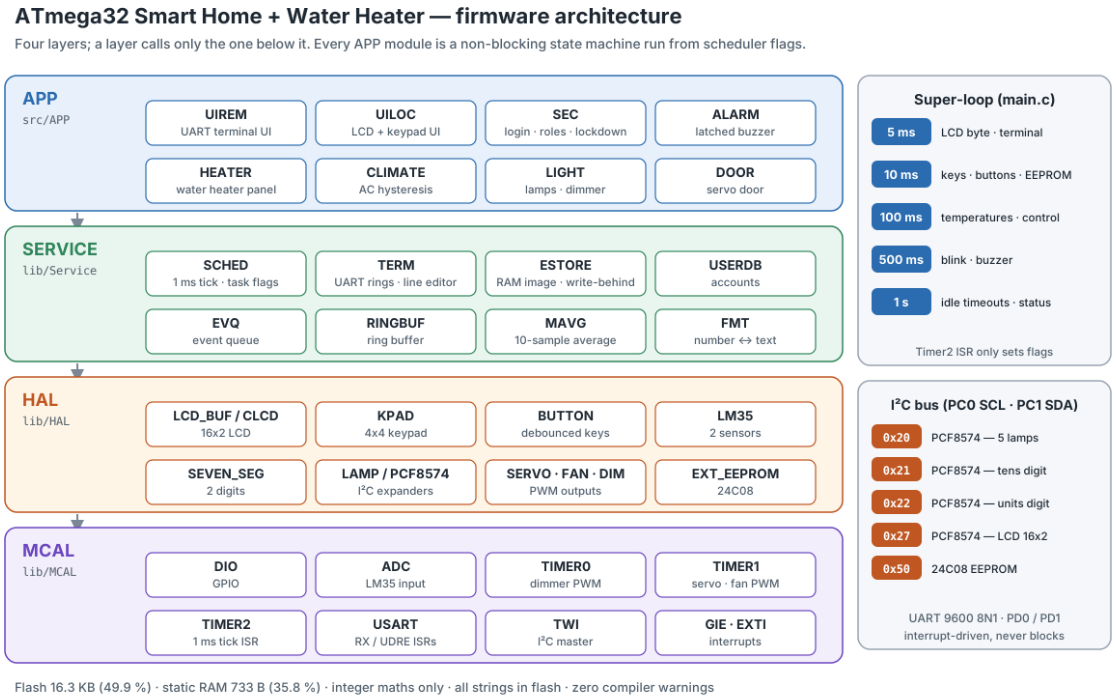


Buses: I²C + UART	4
Keypad 4×4	8
Water-heater panel	7
Smart-home I/O	5
Reserved (LDR, PIR)	2
Spare — freed by I²C	6

~41 I/O lines fit because five devices share one I²C bus

ARCHITECTURE

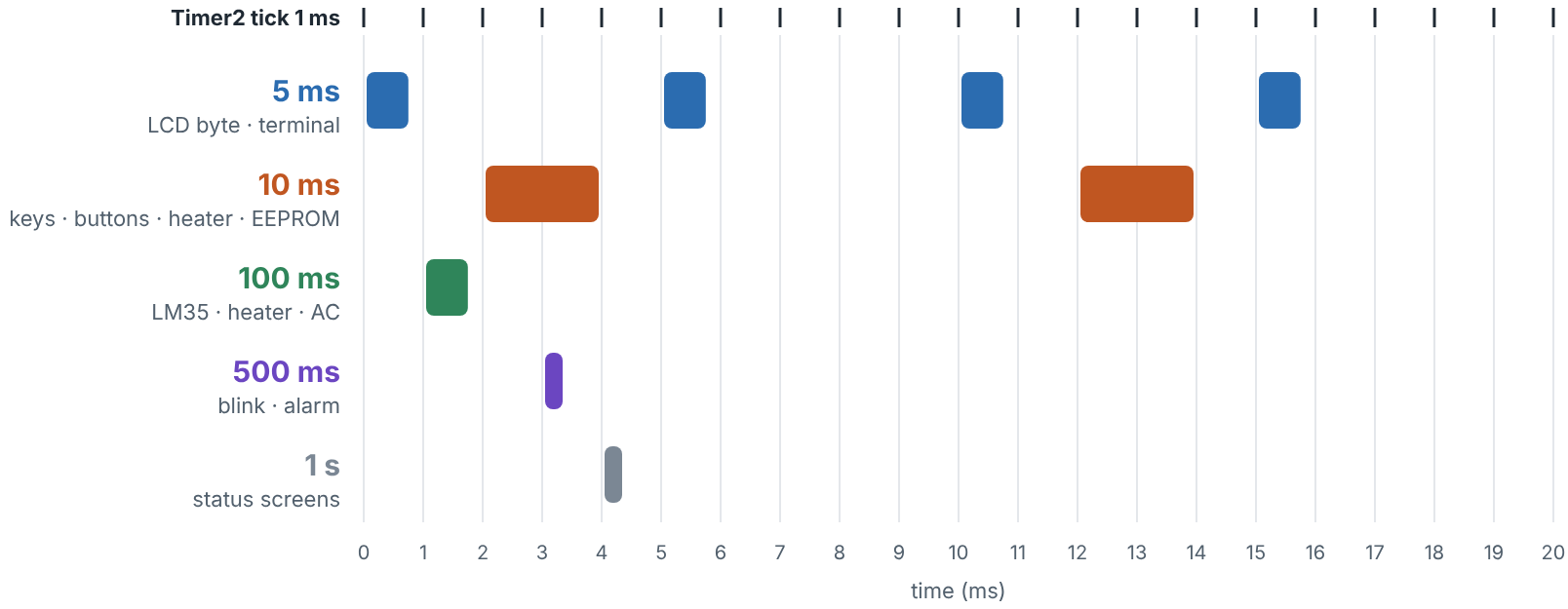
Layered firmware architecture



Four layers; each calls only the one below

SCHEDULER

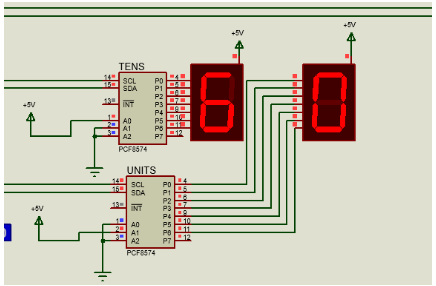
1 ms tick, non-blocking super-loop



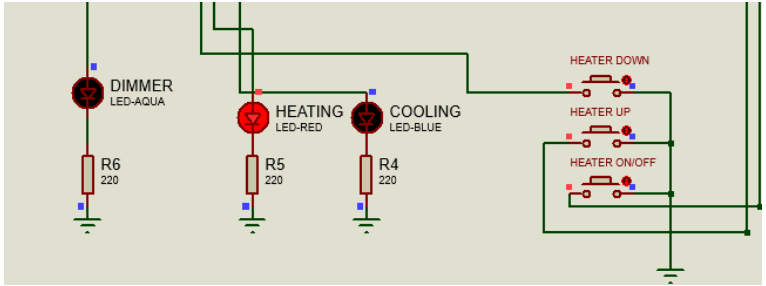
Staggered slots never collide; worst loop pass ≈ 2.2 ms

HEATER

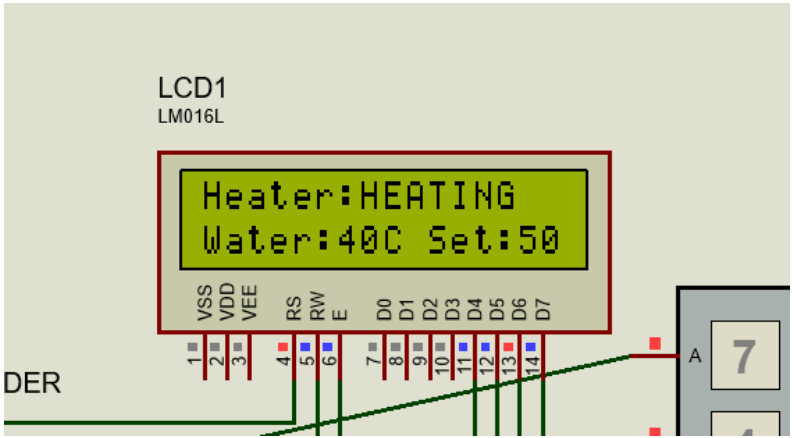
Electric water heater



Water temperature on two I²C digits



Heating / cooling elements and panel buttons

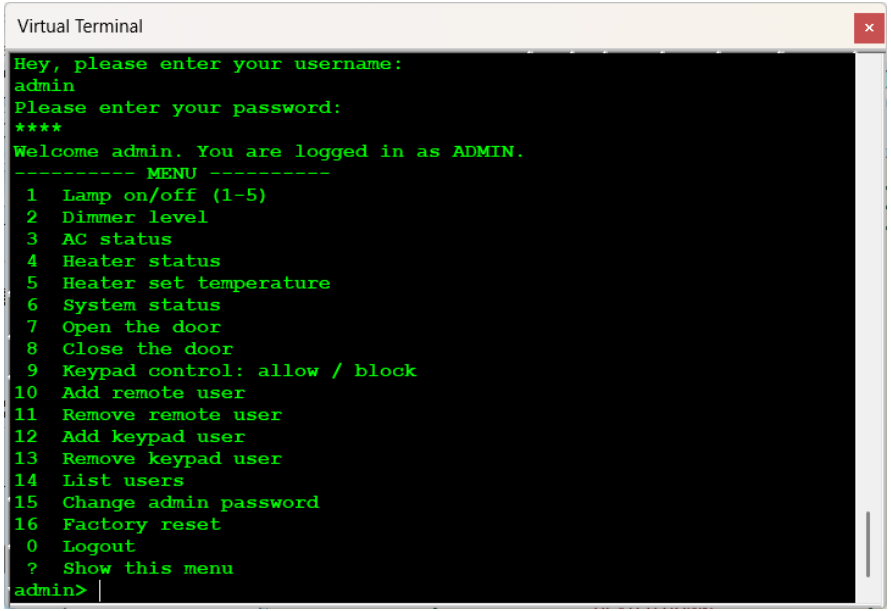


LCD status page while heating

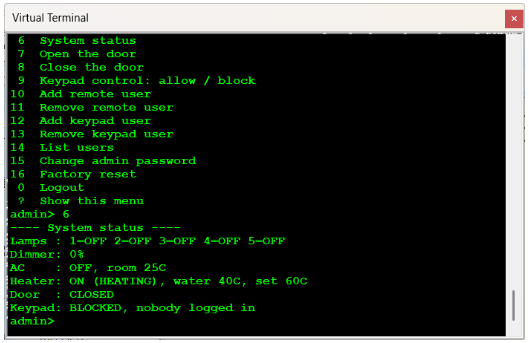
Set 35–75 °C; ±5 °C band on a 10-sample average

INTERFACES

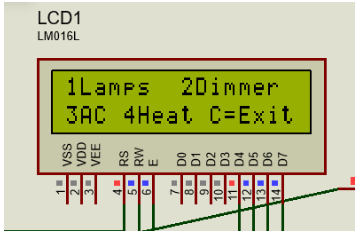
Two user interfaces



UART terminal · admin menu



Live system status

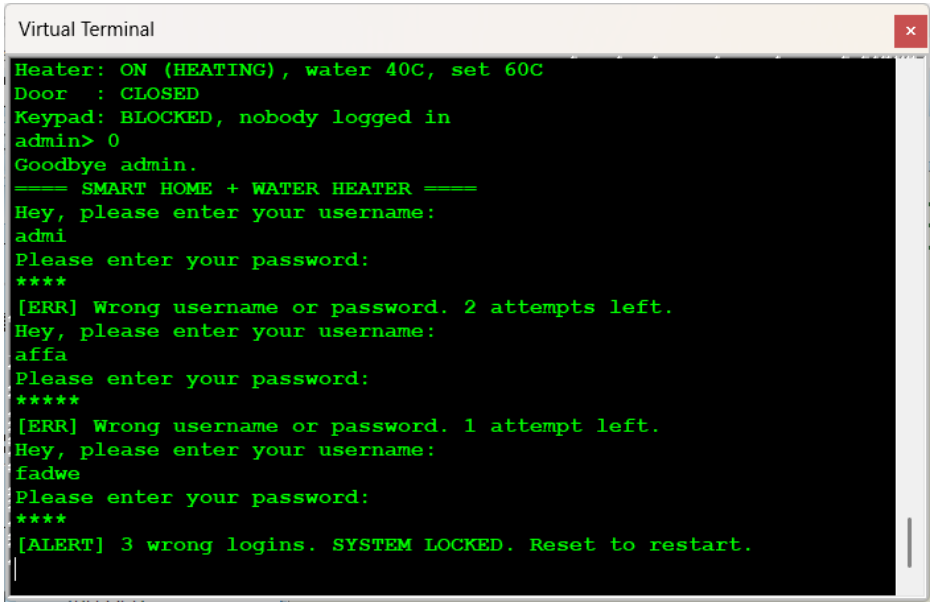


Keypad + LCD menu

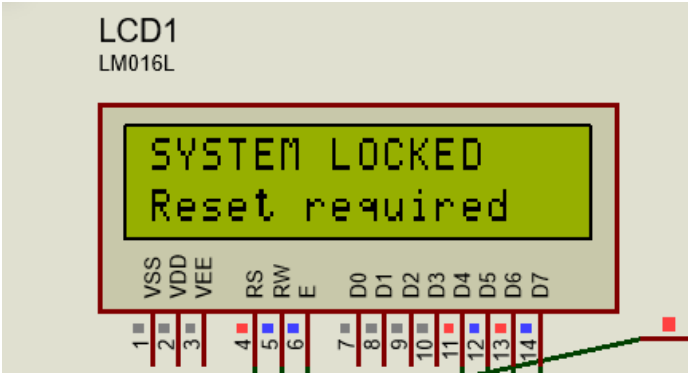
Terminal for admin and users; keypad + LCD for local users

SECURITY

Security and lockdown



Third wrong login in a row

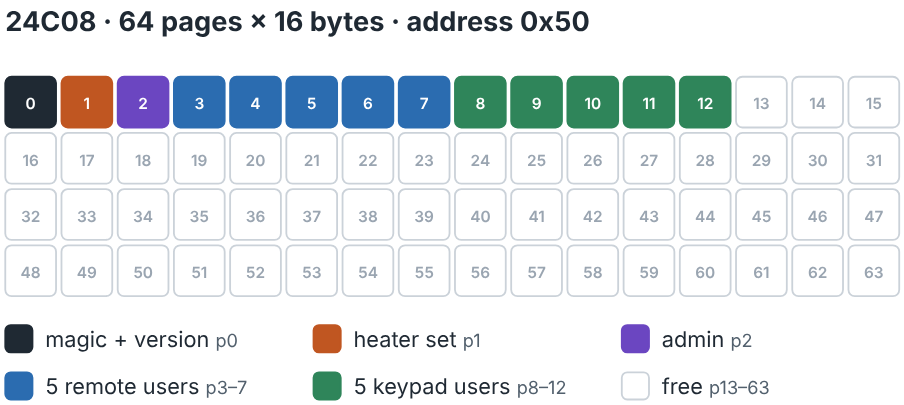


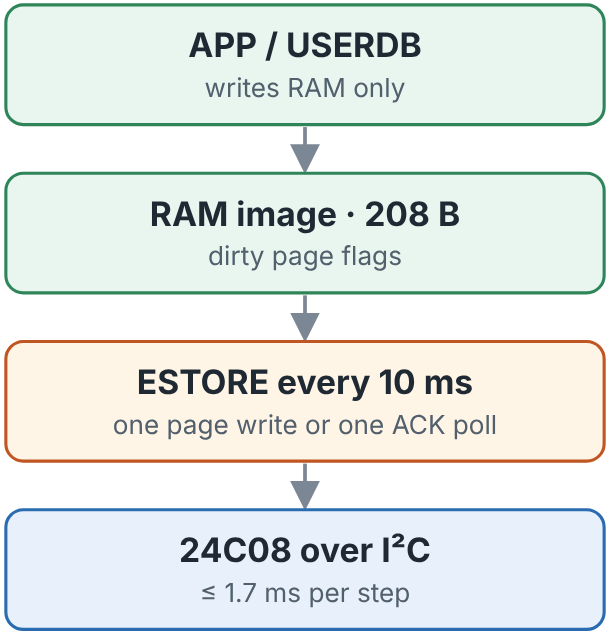
LCD in lockdown

Three wrong logins lock the system until reset

STORAGE

EEPROM storage that never blocks





24C08 over I²C: RAM image plus write-behind

TESTING

Verified layer by layer

ATmega32 SMART HOME + WATER HEATER · PHASE 2 APP · PROTEUS VERIFICATION

APP Layer — Part A Test Sequence

ALL STEPS PASSED

Modules: LIGHT · DOOR · CLIMATE · HEATER · ALARM · SEC

Firmware test_app (branch phase2-app) · Proteus run 2026-10-04 · Tester: Abdelrahman Sherif Medhat

9	48	6	0
test blocks	checked steps	APP modules	failures

APP Part A · 48 steps

ATmega32 SMART HOME + WATER HEATER · PHASE 2 APP · PROTEUS VERIFICATION

APP Layer — Part B Test Sequence

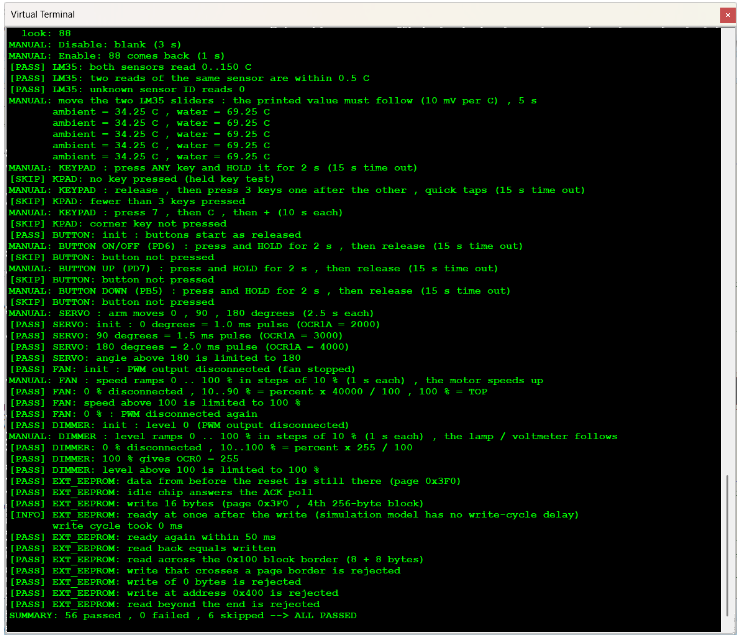
ALL STEPS PASSED

Modules: UILOC (LCD + keypad) · UIREM (UART terminal) · full super-loop, all 8 APP modules

Firmware test_app (branch phase2-app) · Proteus run 2026-10-04 · Tester: Abdelrahman Sherif Medhat

13	82	8	0
test blocks	checked steps	APP modules live	failures

APP Part B · 82 steps

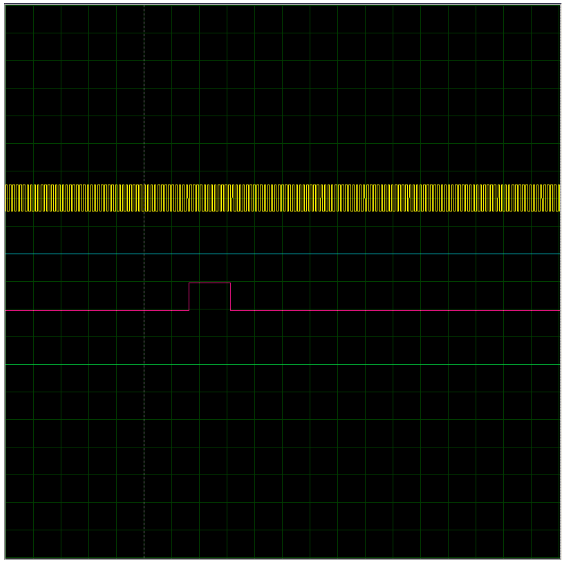


HAL layer test over UART

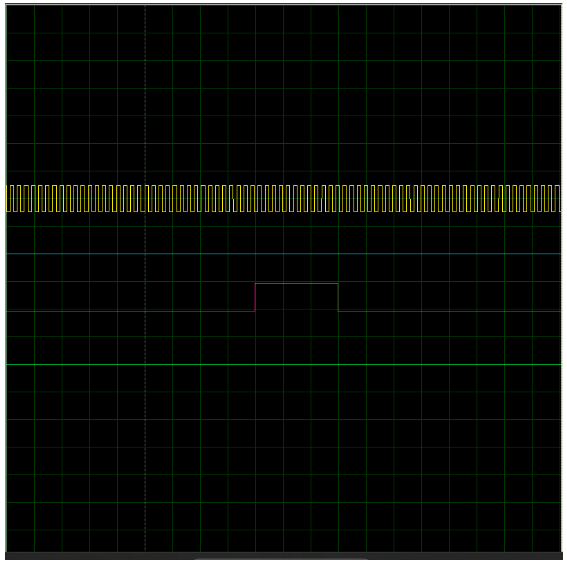
APP layer: 130 / 130 steps passed in Proteus

EVIDENCE

Signals on the scope and the bus



Servo pulse (C) · dimmer PWM (A)



Servo pulse after a door command

```
449.277 s 449.277 s S 4E A 3D A 39 A 5D A 59 A P
449.282 s 449.282 s S 4E A 3D A 39 A 0D A 09 A P
452.117 s 452.117 s S 4E A 3D A 39 A 0D A 09 A P
452.122 s 452.122 s S 4E A 8C A 88 A 0C A 08 A P
452.127 s 452.127 s S 4E A 4D A 49 A 8D A 89 A P
452.132 s 452.132 s S 4E A 6D A 69 A 5D A 59 A P
452.137 s 452.137 s S 4E A 6D A 69 A 1D A 19 A P
452.142 s 452.142 s S 4E A 7D A 79 A 4D A 49 A P
452.147 s 452.147 s S 4E A 6D A 69 A 5D A 59 A P
452.152 s 452.152 s S 4E A 7D A 79 A 2D A 29 A P
452.157 s 452.157 s S 4E A 3D A 39 A AD A A9 A P
452.162 s 452.162 s S 4E A 4D A 49 A FD A F9 A P
452.167 s 452.167 s S 4E A 4D A 49 A 6D A 69 A P
452.172 s 452.172 s S 4E A 4D A 49 A 6D A 69 A P
452.177 s 452.177 s S 4E A 2D A 29 A 0D A 09 A P
452.182 s 452.182 s S 4E A 8C A 88 A CC A C8 A P
452.187 s 452.187 s S 4E A 2D A 29 A 0D A 09 A P
452.192 s 452.192 s S 4E A 2D A 29 A 0D A 09 A P
452.197 s 452.197 s S 4E A 2D A 29 A 0D A 09 A P
452.202 s 452.202 s S 4E A 2D A 29 A 0D A 09 A P
452.207 s 452.207 s S 4E A CC A C8 A 0C A 08 A P
452.212 s 452.212 s S 4E A 5D A 59 A 7D A 79 A P
452.217 s 452.217 s S 4E A 6D A 69 A 1D A 19 A P
452.222 s 452.222 s S 4E A 7D A 79 A 4D A 49 A P
452.227 s 452.227 s S 4E A 6D A 69 A 5D A 59 A P
452.232 s 452.232 s S 4E A 7D A 79 A 2D A 29 A P
452.237 s 452.237 s S 4E A 3D A 39 A AD A A9 A P
452.242 s 452.242 s S 4E A 3D A 39 A 4D A 49 A P
452.247 s 452.247 s S 4E A 3D A 39 A 0D A 09 A P
452.252 s 452.252 s S 4E A 4D A 49 A 3D A 39 A P
452.257 s 452.257 s S 4E A 2D A 29 A 0D A 09 A P
452.262 s 452.262 s S 4E A 5D A 59 A 3D A 39 A P
452.267 s 452.267 s S 4E A 6D A 69 A 5D A 59 A P
452.272 s 452.272 s S 4E A 7D A 79 A 4D A 49 A P
452.277 s 452.277 s S 4E A 3D A 39 A AD A A9 A P
452.282 s 452.282 s S 4E A 3D A 39 A 6D A 69 A P
```

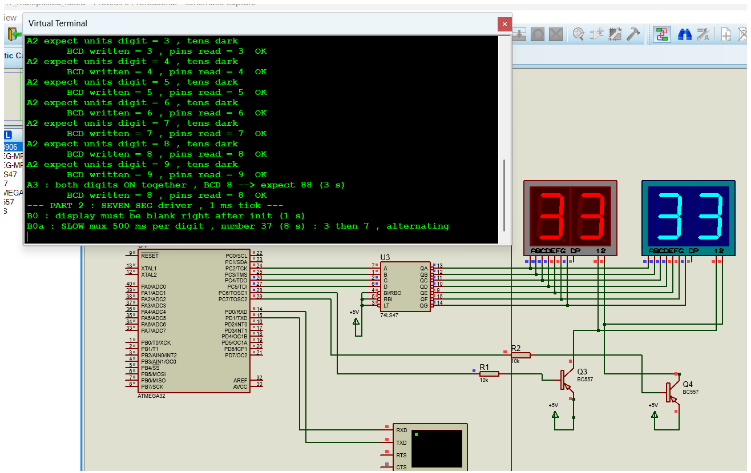
I²C debugger: LCD at 0x27 (0x4E)

PWM timing and I²C traffic checked in the simulator

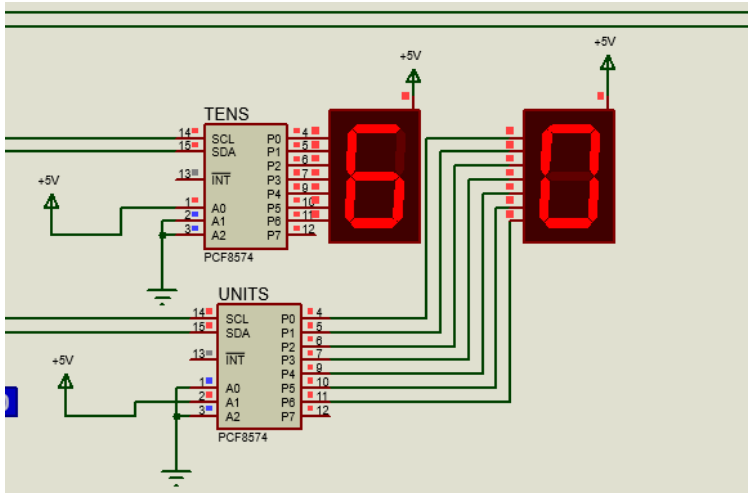
DECISION D-20

The 7-segment redesign

v1 · 7447, multiplexed · failed



v2 · one PCF8574 per digit · final



Multiplexing was unreliable in Proteus; one I²C expander per digit fixed it

BUILD

Results

Flash

49.9 %

16 344 of 32 768 bytes

Static RAM

35.8 %

733 of 2 048 bytes

Compiler warnings

0

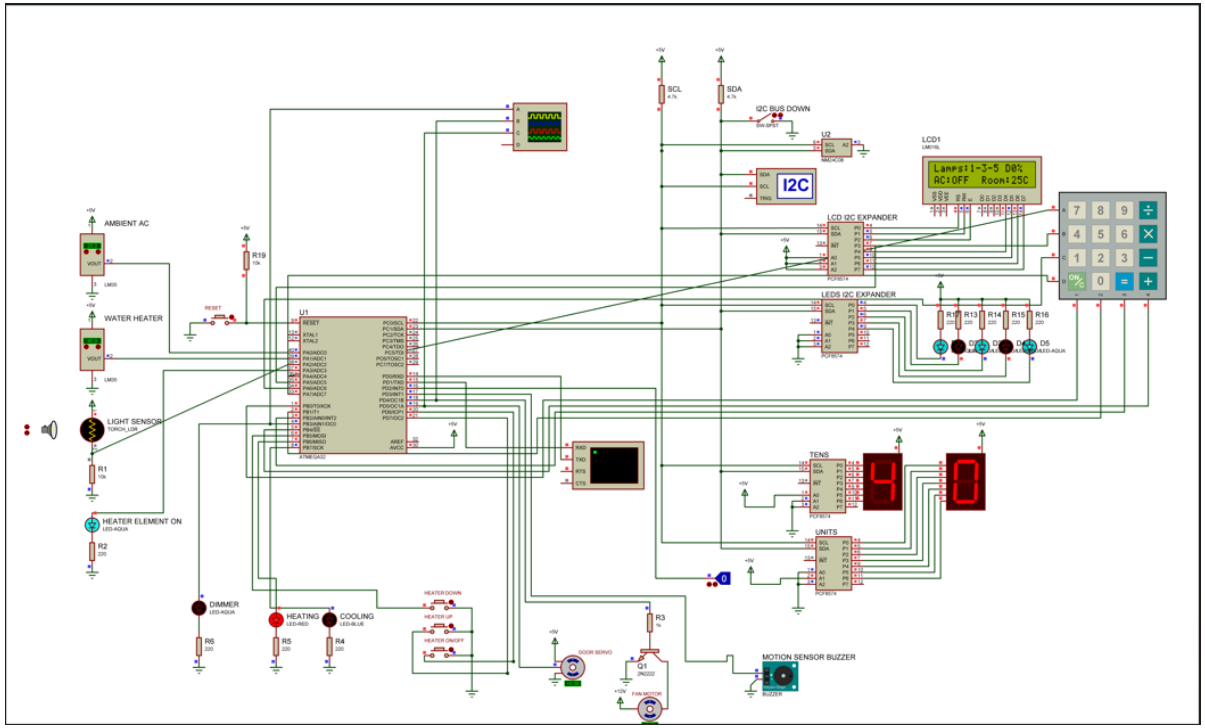
app · test_base · test_mcal · test_hal · test_service · test_app

all six builds with -Wall

Integer maths only; all text kept in flash

LIVE

Demo



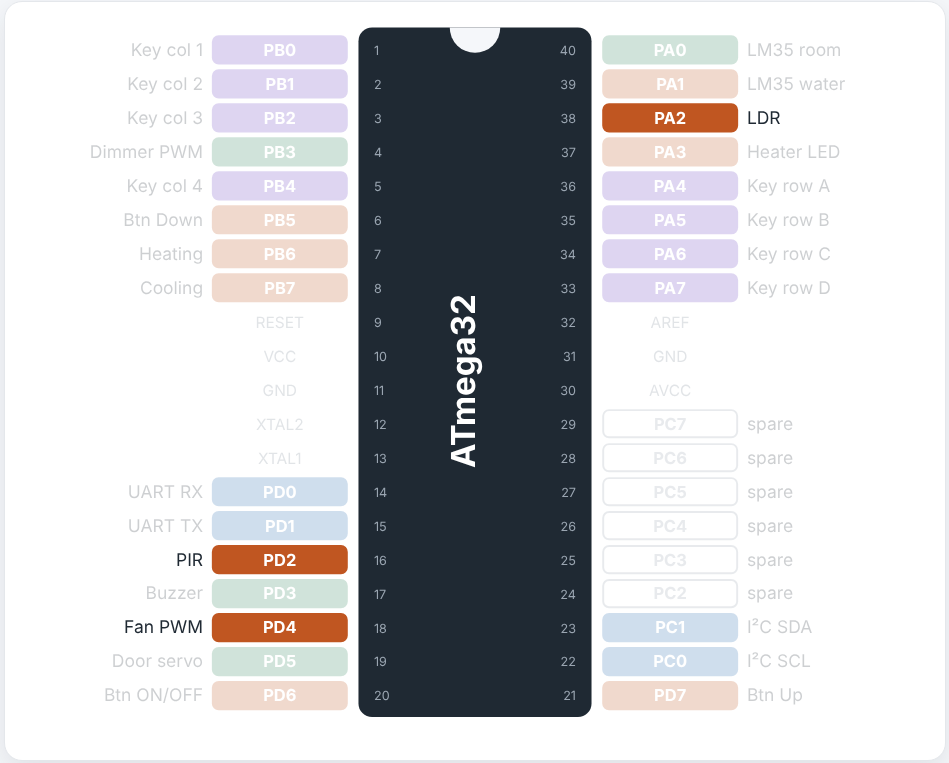
Scan for the demo video

youtu.be/iW8o4F8znDA

Full system demo in Proteus

NEXT

Future work



PD4 · Timer1 OC1B

AC fan speed ramp

PD2 · INT0

PIR motion alarm

PA2 · ADC2

LDR light sensor

Pins already reserved for the next version

Thank you

Abdelrahman Sherif Medhat

Project firmware, simulation and documentation

Course drivers (MCAL/HAL base set): **Eng. Eslam Hefny**, AMIT Learning

github.com/abdelrahmansherifmedhat1134/embedded-diploma